# NAGALAND LITERACY AND NUMERACY FEST 2026

## Micro-Improvement Project (Numeracy Teacher) Grade 6

## Chapter 6 — Perimeter and Area

### Why this Micro-Improvement Project (MIP)?

Across Nagaland, students already live inside the perimeter and area every day: the bamboo fence around a kitchen garden, the terraced boundary of a paddy field, the mat woven for the courtyard, the football ground marked out before a match, or the school. This Micro-Improvement Project (MIP) helps students notice these everyday boundaries and surfaces, name them mathematically, and move from measuring by walking, string, and counting squares toward stating and using formulas for the perimeter and area of a rectangle and a square. It builds directly on the measurement foundations from earlier MIP cycles, and prepares students to reason with shapes rather than only calculate with numbers.

**Skill in Focus:** Perimeter and Area Rectangles and Squares

**Duration:** 1 week

### Learning Outcome

**Concept of Perimeter and Introduction to Area**

- Introduction and general understanding of perimeter using many shapes; shapes of different kinds with the same perimeter.
- Concept of area, area of a rectangle and a square; counter-examples to different misconceptions related to perimeter and area.
- Perimeter of a rectangle and its special case, a square deducing the formula for perimeter of a rectangle and then a square through pattern and generalisation.

### Key Concepts

- Demonstrates the concept of perimeter using different shapes.
- Explains the concept of area.

- Deduces the formula of the perimeter and the area for a rectangle and then a square through pattern and generalisation.
- Provides counter-examples to different misconceptions related to perimeter and area.

**Contempency Addressed C-3.2**: *"Outlines the properties of lines, angles, triangles, quadrilaterals, and polygons and applies them to solve related problems."*

## Detailed Tasks

| Task-1 | Walking the Boundary — What is Perimeter? |
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| **Objective** | Build an intuitive, whole-body understanding of perimeter as the distance around a shape, using the school ground or a real local boundary. |
| **Teacher's Note** | Outdoor. Take students to the school field, courtyard, or garden bed. If ground space is limited, use the classroom floor or veranda. |
| **Duration** | 30 minutes |

**Materials Needed**

Ball of string or rope • Chalk • Measuring tape or a bamboo stick of known length • Notebook and pencil

### Activity Steps

1. Take students to a familiar boundary: the school football/playground ground, a kitchen garden, or the classroom and ask: 'If we wanted to build a bamboo fence around this space, what would we need to know first?'
2. In pairs, students walk the full boundary of the space, counting footsteps or laying string along the edge, and mark the start and end point with chalk.
3. Students measure the string length used (or footsteps taken) and record it as the perimeter of that space in their notebook.
4. Repeat with two or three different-shaped spaces of similar size (a triangular flower bed, a rectangular classroom, an irregular patch) and compare the boundary lengths.
5. Close with a class discussion: 'Two shapes looked very different. Did they need the same length of fencing or a different length? What does that tell us about the perimeter?'

**Concept Built:** *Perimeter as the total distance around the boundary of a shape.*

| Task-2 | Shape Shifters — Same Perimeter, Many Shapes |
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| **Objective** | Show that many different shapes can share exactly the same perimeter, using a fixed loop of string or rope. |
| **Teacher's Note** | Outdoor or Classroom (open floor space). |
| **Duration** | 30 minutes |

**Materials Needed**

A loop of string or rope of fixed length (e.g. 4 metres) per group • Chalk • Notebook

**Activity Steps**

1. Give each group a loop of rope of exactly the same length, tied at the ends.
2. Ask each group to lay the loop on the ground to form a rectangle, and trace its outline with chalk.
3. Without cutting or adding rope, ask groups to reshape the same loop into a square, a triangle, and an irregular shape, tracing each on the ground.
4. Groups compare their tracings: the shapes look completely different, but every one uses the same length of rope.
5. Discuss as a class: 'Since all these shapes were made from the same rope, what do we know is the same about them? What looks different?' Introduce the word 'perimeter' formally, and note that the space enclosed inside each shape looks different — this sets up the idea of area in Task-3.

**Concept Built:** *Shapes of different kinds can have the same perimeter, even though they look and enclose different amounts of space.*

| Task-3 | Covering the Courtyard — Introducing Area |
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| **Objective** | Introduce area as the amount of surface covered inside a shape, using unit squares, and connect it to the area of a rectangle and a square. |
| **Teacher's Note** | Classroom. |
| **Duration** | 35 minutes |

**Materials Needed**

Graph paper or paper with a hand-drawn grid • Small tiles, leaves, or bottle caps (as unit squares) • Crayons or sketch pens

**Activity Steps**

1. Show students a woven mat or a Naga shawl folded into a rectangle and ask: 'If we wanted to know how much cloth or floor this covers, is walking the edge (perimeter) enough? What else could we do?'
2. Give each student a grid sheet. Ask them to draw a rectangle covering, say, 4 rows and 6 columns of the grid, and count the number of full unit squares inside it.
3. Students record: length of rectangle = 6 units, breadth = 4 units, number of unit squares (area) = 24.
4. Repeat with a square of side 5 units on the grid, counting the unit squares to find its area.
5. Alternatively, for classrooms without grid paper, students can lay small leaves or bottle caps to fully cover a drawn rectangle on the floor or desk, then count how many were used, arriving at the same idea by counting covering pieces instead of grid squares.
6. Discuss: 'What did counting the squares tell us that walking the boundary did not?' Introduce the term 'area' as the amount of surface covered inside a shape.

**Concept Built:** *Area as the surface covered inside a shape, first found by counting unit squares, for rectangles and squares.*

| Task-4 | Busting the Myths — Perimeter vs Area |
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| **Objective** | Confront and correct common misconceptions that a bigger perimeter always means a bigger area, and that shapes with the same perimeter must have the same area (or vice versa), using worked counter-examples. |
| **Teacher's Note** | Classroom. |
| **Duration** | 30 minutes |

**Materials Needed**

Grid/graph paper • Chart paper for the class 'Myth or Fact?' board • Sketch pens

**Activity Steps**

1. Present two rectangles on the board: one 2 units by 6 units, and one 4 units by 4 units (a square). Ask students to calculate the perimeter of each both equal 16 units.
2. Ask students to now calculate the area of each: 2 × 6 = 12 square units, and 4 × 4 = 16 square units. Students discover the areas are different even though the perimeters are the same.

3. Present a second pair a rectangle 3 units by 8 units, and one 4 units by 6 units and ask students to check that both have the same area (24 square units) but different perimeters (22 units and 20 units).
4. In groups, students write one 'Myth' (a common wrong belief, e.g. 'A bigger fence around a field always means a bigger field') and one 'Fact' (their correct counter-example with numbers), and pin it to a class 'Myth or Fact?' chart.
5. Close by asking each group to explain their counter-example aloud to the class in their own words.

**Concept Built:** *Same perimeter does not guarantee the same area, and same area does not guarantee the same perimeter — demonstrated through worked counter-examples.*

| Task-5 | Cracking the Code — Deducing the Perimeter Formula |
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| **Objective** | Guide students to discover the formula for the perimeter of a rectangle, and then its special case, the square, through pattern recognition rather than being told the formula directly. |
| **Teacher's Note** | Classroom. |
| **Duration** | 35 minutes |

**Materials Needed**

Grid/graph paper • Ruler • Notebook for recording the pattern table

**Activity Steps**

1. Students draw four different rectangles on grid paper and, for each, count the length, the breadth, and the perimeter by adding all four sides. Record these in a three-column table: Length, Breadth, Perimeter.
2. Ask students to look for a pattern connecting length and breadth to the perimeter without adding all four sides one by one. Guide them toward noticing that perimeter = length + breadth + length + breadth.
3. Help students simplify this to Perimeter = 2 × (Length + Breadth), and test the formula against their table to confirm it matches every row.
4. Ask: 'What happens to this formula when the length and breadth are equal — like a square courtyard?' Students substitute length = breadth = side, simplifying 2 × (side + side) to Perimeter = 4 × side.
5. Students verify the square formula using a real example, such as a square bamboo enclosure of side 3 metres, calculating the fencing required.

**Concept Built:** *Perimeter of a rectangle, P = 2 × (length + breadth), and its special case for a square, P = 4 × side, deduced through pattern and generalisation.*

| Task-6 (Homework) | Measure My World |
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| **Objective** | Apply perimeter and area concepts to a real object or space at home, connecting classroom learning to daily life in students' own villages and households. |
| **Teacher's Note** | Homework. To be done at home with family, and discussed in class the next day. |
| **Duration** | 20 minutes |

**Materials Needed**

Measuring tape, string, or a stick of known length • Notebook and pencil

**Activity Steps**

1. Ask each student to choose one rectangular or square object or space at home: a mat, a courtyard, a garden bed, a bamboo enclosure, a window, or a field corner.
2. Students measure its length and breadth using string, a stick, or a tape, and record both measurements in their notebook.
3. Using the formulas from Task-5, students calculate the perimeter and, using the method from Task-3, the area of the chosen object or space.
4. Students write one sentence explaining what the perimeter tells them (e.g. how much fencing or border material is needed) and one sentence explaining what the area tells them (e.g. how much land, mat, or floor is covered).
5. The next day, students share their chosen object and calculations with a partner or the class.

**Concept Built:** *Transfers perimeter and area formulas to a self-chosen, real-life object, strengthening the link between classroom mathematics and daily life.*

## Culminating Showcase

| Showcase | Nagaland Space Designers — Final Project |
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| **Objective** | Bring together every concept from Tasks 1–6 into one creative, real-world design project that groups plan, measure, and present. |

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| **Teacher's Note** | Combined Classroom and Outdoor project, spread across 2–3 sessions after Task-6, culminating in a class exhibition. |
| **Duration** | 30–35 minutes per session (2–3 sessions) |

**Materials Needed**

Chart paper or cardboard • Ruler, pencil, crayons/sketch pens • String or measuring tape • Locally available materials for models (bamboo sticks, leaves, small stones) — optional

**Activity Steps**

1. In groups of 4–5, students choose a space to design that is meaningful in their community for example, a kitchen garden, a playground or football court, a paddy field terrace, a classroom, or a stall layout for a village festival.
2. Groups decide the length and breadth of each rectangular or square section of their design (e.g. planting beds, a fenced boundary, a seating area) and draw it to scale on chart paper or grid paper.
3. Groups calculate the total perimeter of the outer boundary (to estimate fencing or border material needed) and the total area of each section (to estimate land, planting, or flooring material needed), showing their working clearly on the chart.
4. Groups identify and label at least one place in their design where they deliberately used the idea from Task-4 for instance, showing that a smaller-perimeter section can still have a larger area, or vice versa.
5. In the final session, each group presents its design in a short class exhibition, explaining the shapes used, the perimeter and area calculations, and one interesting or surprising discovery from their design process.

**Concept Built:** *Independent application of perimeter, area, formulas, and misconception-awareness in one integrated, creative design task.*

***By the end of these tasks, teachers and students should have moved from walking boundaries and counting squares to confidently naming, calculating, and designing with perimeter and area culminating in an original, locally-grounded design that each group can showcase with pride.***

**Reflect on Students' Understanding**

Observe 5–6 students each week during the tasks above. No written test is needed to watch what students do with string, grids, and shapes, and listen to how they explain their thinking.

| Student Name | Explains Perimeter | Computes Area | Derives Formula | Corrects a Misconception | Needs Support |
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*Scale (for each column):* ☐ *Not Yet* ☐ *Sometimes* ☐ *Consistently*

**Picture Guide**